



N4523A26R0809

**Radar Depot Mast Thermal Spray Service
Contract Statement of Work (SOW)**

**Multiple Award Contract-Indefinite Delivery
Indefinite Quantity (MAC-IDIQ)**

Dated: 27 FEB 2026

STATEMENT OF WORK (SOW)

1. GENERAL INFORMATION:

- 1.1. GENERAL: This is a non-personnel services contract to provide thermal spray repair technical services to submarine radar mast tubes upon approved CFR. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.
- 1.2. DESCRIPTION OF SERVICES/INTRODUCTION: The contractor shall provide all personnel, facilities, equipment, supplies, transportation, tools, materials, supervision, other items and non-personal services necessary to perform thermal spray repair services to submarine radar mast tubes as defined in this Statement of Work (SOW), except for those items specified as government furnished material, property and services. The contractor shall perform to the standards in this contract.
- 1.3. HIGH RISK JUSTIFICATION: Service supports the restoration of SSBN 726 Class, SSN 21 Class, and SSN 774 Class variants of the AN/BPS-16 Radar Mast Sub-Assembly, and SSN 688 Class, SSN 688I Class, and SSBN 726 Class variants of the AN/BPS-15 Radar Mast Sub-Assembly, that directly affect the successful completion of submarine availabilities at US Navy shipyards, regional maintenance centers, and refit facilities. Delays to the timely procurement and accomplishment of services directly affect availability schedules and has potential to delay the underway of combatant and strategic submarine units.

****SUBSAFE****

****LEVEL 1****

- 1.4. BACKGROUND: The Trident Refit Facility Bangor Radar Depot performs restoration of mechanical radar components installed on the various classes of submarines for US Navy Submarine Projects at Shipyards and maintenance activities, in addition to supporting Navy Supply System NSN Depot Level Repairable (DLR) asset restorations. During restoration of radar mast components, the outer surfaces of radar mast tubes typically require surface finish refurbishment due to wear and corrosion defects that do not meet inspection criteria of the applicable NAVSEA Maintenance Standard document, which requires NAVSEA certified thermal spray deposition processes and facilities be used to affect refurbishment of the radar mast tube surfaces.
 - 1.4.1. Radar mast tubes must maintain their SUBSAFE/Level 1 certifications at all times, and repairs must be performed by NAVSEA certified SUBSAFE entities listed in NAVSEANOTE 5000, Ser. 07/056 DTD 29 OCT 2024. Barring available qualified SUBSAFE activities to affect thermal spray repair services, TRFB, a NAVSEA qualified "SUBSAFE Certifying Activity", will provide SUBSAFE oversight, via a SUBSAFE Memorandum of Agreement (MoA) in accordance with (IAW) TRFB Engineering Memorandums 410/214M-21 Rev-A, Reference 2.1, and 410/215M-21 Rev-B, Reference 2.2, for the repair process required by contract for SUBSAFE/Level 1 work performed by contractors not previously certified to perform SUBSAFE work by NAVSEA.

1.5. SCOPE:

- 1.5.1. TRFB's Radar Depot performs restoration of SSN 774 Class, SSN 21 Class, and SSBN 741-743 Class submarine variants of the AN/BPS-16 radar mast sub-assembly. Each AN/BPS-16 radar mast sub-assembly variant is comprised of three distinct radar mast tubes: one Outer Housing, one Radar Mast, and one Rotary Tube detailed in Table 1; Ship Class AN/BPS-16 Mast Variants.

Table 1; Ship Class AN/BPS-16 Mast Variants

<u>AN/BPS-16 MAST VARIANT</u>	<u>RADAR MAST TUBE</u>	<u>CAGE: 03956 P/N & Detail DWG</u>	<u>CAGE 03956 DWG & ITEM#</u>	<u>NATIONAL STOCK NUMBER (NSN)</u>
SSBN726 CLASS	INNER MAST	1978731	PL1978837-2; IT 22	5985-01-633-7511
	ROTARY TUBE	1978730	PL1978837-2; IT 21	5840-01-610-0361
	OUTER HOUSING	1980470	PL1978837-2; IT 23	5985-01-633-7504
SSN21 CLASS	INNER MAST	1978732	PL1978837-3; IT 22	5985-01-633-7512
	ROTARY TUBE	1978730	PL1978837-3; IT 21	5840-01-610-0361
	OUTER HOUSING	1981061	PL1978837-3; IT 23	5985-01-633-7514
SSN774 CLASS	INNER MAST	1978732	PL1978837-4; IT 22	5985-01-633-7512
	ROTARY TUBE	1978730	PL1978837-4; IT 21	5840-01-610-0361
	OUTER HOUSING	1980470	PL1978837-4; IT 23	5985-01-633-7504

- 1.5.1.1. AN/BPS-16 radar mast tubes requiring repair of surfaces IAW sheets 1 through 4 of Enclosure 3 of Reference 2.2. Reference 2.2, Enclosure (2), paragraph 1 specifies refurbishment of the outer surfaces of the mast tubes using thermal spray processes identified in Reference 2.3.

- 1.5.2. TRF-B's Radar Depot performs restoration of SSN 688 Class, SSN 688I Class, and SSBN 726 Class submarine variants of the AN/BPS-15 radar mast sub-assembly. Each AN/BPS-15 radar mast variant is comprised of one radar mast tube detailed in Table 2; Ship Class AN/BPS-15 Mast Variants.

Table 2; Ship Class AN/BPS-15 Mast Variants

<u>AN/BPS-15 MAST VARIANT</u>	<u>RADAR MAST TUBE</u>	<u>CAGE 53711 DWG & ITEM#</u>	<u>NATIONAL STOCK NUMBER (NSN)</u>
SSBN 726 CLASS	MAST	7226178; IT 3	5985-01-475-1217 (Mast) N/A (Tube)
SSN 688I CLASS	MAST	6640200; IT 3	5985-01-335-9763 (Mast) 5985-01-366-6928 (Tube)
SSN 688 CLASS	MAST	6640198; IT 3	5985-01-335-9762 (Mast) 5985-01-366-6929 (Tube)

- 1.5.2.1. AN/BPS-15 radar mast tubes requiring repair of surfaces IAW sheets 1 through 4 of Enclosure 3 of Reference 2.1. Reference 2.1, Enclosure (2), paragraph 1 specifies refurbishment of the outer surfaces of the mast tubes using thermal spray processes identified in Reference 2.3.

- 1.6. TYPE OF CONTRACT: Multiple Award Contract for Indefinite Delivery Infinite Quantity (MAC-IDIQ).

1.6.1. US Navy Submarine SSN 774, SSN 21, and SSBN 741-743 class maintenance plans mandate scheduled replacement of AN/BPS-16 Radar Mast Sub-Assys that drives refurbishment of "turned in" DLR NSN assets by the Radar Depot. Additionally, in-service/operational degradation of AN/BPS-15 and AN/BPS-15 Radar Mast Sub-Assys drive DLR NSN asset consumption and unanticipated/unscheduled refurbishment of DLR NSN assets by the Radar Depot. Since fiscal year 2019, these factors have led to a semi-regular demand signal to the thermal spray industrial base such to attract interest among all the NAVSEA qualified sources of supply. Contractor lead-times and period-of-performance durations, coupled with Technical Receipt Inspection (TRI) and SUBSAFE data acceptance process experienced during prior Firm Fixed Price (FFP) contracts issued in Fiscal Years 2019, 2021, 2022, 2023, & 2025 necessitate a change in contracting strategy when factoring in the scheduled demand signal alone, but include the unanticipated demand signal and the TRFB Radar Depot will be unable to meet mission requirements.

1.6.1.1. TRFB command level discussions desire the flexibility that a MAC-IDIQ contracting vehicle affords over a five year term, with a five-year ordering period. This contracting vehicle is preferable over continuing to issue FFP contracts that have supported the Radar Depot since fiscal year 2019, which have contributed to maintenance backlogs and excessively long Repair Turn Around Times (RTAT) of DLR NSN assets, in addition to increasing TRFB's administrative burden from submitting multiple requests for service contracts and generating the supporting documentation for each request.

1.6.1.2. Procurement of thermal spray refurbishment service for each radar mast tube will be a Job Order specified by Contract Line Item Number (CLIN), listed initially or upon award of each successive CLIN added after initial contract award. Each CLIN will specify the AN/BPS-16 or AN/BPS-15 mast tube description, drawing and item, and the mast tube's unique identifying serial number.

1.6.1.3. Typical procurement of thermal spray refurbishment services of AN/BPS-16 radar mast sub-assemblies includes one of each radar mast tube type listed in Table 1 for the specified radar mast variant.

1.6.1.3.1. TRFB Radar Depot work requirements and emergent submarine fleet work requests may necessitate procurement of thermal spray refurbishment services of less than a full set of three AN/BPS-16 radar mast tubes, or any combination of one Outer Housing, one Radar Mast, and one Rotary Tube listed in Table 1.

1.6.1.4. Typical procurement of thermal spray refurbishment services of AN/BPS-15 radar mast sub-assemblies include one mast tube listed in Table 2 for the specified radar mast variant.

1.7. PERIOD OF PERFORMANCE: 3 AUG 2026 to 2 AUG 2031

1.7.1. The contract shall include a five-year ordering period through 2 AUG 2031.

1.8. LOCATION OF WORK: Vendor Facility; CONUS

1.9. TECHNICAL CLARIFICATION: All technical clarifications will be coordinated through the Contracting Officer's Representative (COR).

1.10. GOVERNMENT POINT OF CONTACT: Contracting Officer's Representative (COR) Identified in contract.

1.11. SHIPPING AND DELIVERY: Coordinate delivery and shipment of Government Furnished Equipment (GFE) with the COR. See below for clarifying delivery and shipping information:

1.11.1. Freight and shipping, by traceable means, of, up to, a set of three AN/BPS-16 radar mast tubes, or one AN/BPS-15 radar mast tube, from TRFB to the contractor is the responsibility of, and funded by, TRFB. TRFB will provide the shipment tracking identification data to the contractor within seventy-two hours of shipping carrier pick-up from TRFB.

1.11.2. Freight and shipping, by traceable means, of refurbished AN/BPS-16 radar mast tubes, or refurbished AN/BPS-15 radar mast tube, from the contractor to TRFB is the responsibility of, and funded by, the contractor. The contractor shall provide the shipment tracking identification data to the TRFB COR within seventy-two hours of shipping carrier pick-up from the contractor.

FOB Destination:

TRIDENT Refit Facility Bangor

7000 Finback Circle

BLDG 7000 Door 12 Receiving

Silverdale, WA 98315-7000

ATTN: Shop 67R Supervisor, PH: (360) 315-1193

1.11.3. To assist with determining return freight shipping costs the dimensions and maximum weight of loaded shipping crates are provided below:

SSN 21/SSN 774 Class AN/BPS-16 radar mast tubes (three or less):

Shipping Crate Dimensions/Weight: 34"W x 26"H x 116"LG/1450 lbs. (approximately)

SSBN 726 Class AN/BPS-16 radar mast tubes (three or less):

Shipping Crate Dimensions/Weight: 34"W x 26"H x 176"LG/1750 lbs. (approximately)

SSBN 726 Class AN/BPS-15 radar mast tube:

Shipping Crate Dimensions/Weight: 17"W x 21"H x 243"LG/670 lbs. (approximately)

SSN 688/SSN 688I Class AN/BPS-15 radar mast tube:

Shipping Crate Dimensions/Weight: 13"W x 17"H x 172"LG/420 lbs. (approximately)

1.11.4. Delivery to TRFB, within the confines of US Naval Base Kitsap Bangor (NBK-Bangor), requires the delivery carrier driver(s) to have documented proof of US citizenship on their person (no photocopies permitted) to obtain temporary badging at the NBK-Bangor Visitor Control Center (VCC). Alternatively, the delivery carrier driver(s) may possess

pre-issued NBK-Bangor/TRFB badging credentials to enable entry to NBK-Bangor and TRFB.

1.11.5. NBK-Bangor VCC and commercial entry gate operational schedule is Monday through Friday 07:00 to 15:30 and delivery must be made during this period.

2. REFERENCES:

- 2.1. TRF Bangor Engineering Memorandum 410/214M-21 Rev-A, SUBSAFE; IDIQ Thermal Spray BPS-15 Mast Components by Non-SUBSAFE Qualified Contractor [CUI//SP-CTI/SP-EXPT]
- 2.2. TRF Bangor Engineering Memorandum 410/215M-21 Rev-B, SUBSAFE; IDIQ Thermal Spray BPS-16 Mast Components by Non-SUBSAFE Qualified Contractor [CUI//SP-CTI/SP-EXPT]
- 2.3. NAVSEA T9074-AA-GIB-010/1687 dtd 12 JUN 2017; Thermal Spray Process for Naval Ship Machinery and Non-Skid Applications
- 2.4. MIL-STD-1687A; Thermal Spray Processes for Naval Ship Machinery Applications, includes Notice 2 dtd 12 JUN 2017; Notice of Cancellation
- 2.5. NAVSEA T9074-AS-GIB-010/271 Rev-1 dtd 11 SEP 2014; Requirements for Non-Destructive Test Methods
- 2.6. ASNT SNT-TC-1A 2020 Edition, Personnel Qualification and Certification in Nondestructive Testing

3. REQUIREMENTS:

- 3.1. Provide all facilities, tools, parts, equipment, consumables, including hazardous materials, personnel, labor, technical expertise, and Personnel Protection Equipment (PPE) to accomplish requirements.

3.2. SECURITY REQUIREMENTS:

- 3.2.1. Consider this project, and all materials including records, files, documents, and work papers provided to the contractor by the Government and all test results, conclusions and recommendations obtained thereof as Government property.
- 3.2.2. The information shall not be disclosed, copied, modified, used (except in the completion of this contract) or otherwise disseminated to any other person or entity at any time to include, but not limited to, inclusion in any database external to the Government without the Government's expressed written consent.

3.3. TECHNICAL WORK REQUIREMENTS:

- 3.3.1. For consideration under this contract, the contractor's thermal spray facility, quality assurance program, and the thermal spray procedures used in support of these services shall be previously NAVSEA certified IAW paragraphs 5.1 and 5.2 of Reference 2.3. Documented proof thereof shall be provided at solicitation with their proposal for technical review and consideration; see Technical Exhibit 1 Deliverables Schedule.

3.3.1.1. Reference 2.4 is the previous approved military specification for thermal spray processes for naval ship machinery applications, which was superseded by Reference 2.3 as detailed in "Notice 2" of Reference 2.4. Contractors who were previously NAVSEA certified to perform machinery restorative thermal spray processes, along with any previously approved machinery restorative thermal spray procedures under Reference 2.4 are satisfactory for performance and contract award per NAVSEA Thermal Spray In-Service Engineering Agent (ISEA) representatives via email communication.

3.3.2. The contractor shall not use any sub-contractor to perform any work associated with this service, the only exception permitted is the use of a sub-contractor to perform nondestructive liquid dye penetrant testing (NDT PT). When subcontracting NDT PT, all work shall be conducted at the facility of, and under direct supervision of, the primary contractor. All handling of the components (e.g., lifting, positioning, and/or transporting) shall be performed by the primary contractor.

3.3.3. For consideration under this contract, the contractor, or sub-contractor if used, must meet all NAVSEA requirements for NDT PT, to include Reference 2.5 and Reference 2.6. Prior to consideration for award, the contractor, or sub-contractor if used, will be evaluated by the NAVSEA activity, TRFB, to the requirements of Reference 2.5 and Reference 2.6. For this service the TRFB nondestructive testing division, code 630, NDT Level III Examiner will perform the review.

3.3.3.1. The contractor, or sub-contractor if used, shall provide the NDT Vendor Qualification Documentation, items a. through e. listed below, at solicitation with their proposal for technical review and consideration; see Technical Exhibit 1 Deliverables Schedule.

3.3.3.1.1. Contractor's/Sub-Contractor's Written Practice.

3.3.3.1.2. NDT LP procedure meeting requirements of Reference 2.5 Chapter 5.

3.3.3.1.3. NDT Personnel Certification Records meeting requirements of Reference 2.6 paragraph 9.4.

3.3.3.1.4. NDT Level I, Level II, and Level III certification documentation Personnel Certification Records meeting requirements of Reference 2.5 paragraphs: 1.6, 1.6.2.1, 1.6.4, and 1.6.8

3.3.3.1.5. Summary of the Contractor's/Sub-Contractor's NDT surveillance program with evidence of a surveillance within one year of the solicitation date.

3.3.3.1.6. Other documentation regarding the NDT program when requested, to enable approval of solicitation.

3.3.3.2. As determined by the TRFB nondestructive testing division, code 630, NDT Level III Examiner the conduct of an initial, and potential follow-on, site visits may be required to verify compliance with the requirements of Reference 2.5 and Reference 2.6. The contractor shall permit TRFB nondestructive testing

division code 630 personnel access to the work site(s) to observe the performance of the NDT PT to verify NDT PT written practice and NDT PT surveillances comply with Reference 2.5 and Reference 2.6 requirements.

3.3.3.2.1. Prior to conducting any NDT PT, contractor shall provide a minimum seven workday's notification via email to the TRFB COR. Notification is required to enable sufficient time for travel arrangements for the NDT PT performance review site visits.

3.3.3.2.2. Initial and follow-on site visits for NDT PT performance review may not excessively impede or delay the contractor, or subcontractor, such to prevent meeting delivery milestone dates or otherwise extend the period of performance of the contract.

3.3.3.2.3. The TRFB nondestructive testing division, code 630, personnel performing the NDT PT performance review may initiate a stoppage of NDT PT work, for not more than forty-eight hours, to enable identifying and communicating NDT PT certification discrepancies and corrective actions required to enable resumption of NDT PT work.

3.3.4. Upon award of contract and prior to commencement of work on SUBSAFE AN/BPS-16 or SUBSAFE AN/BPS-15 radar mast tubes, the contractor shall permit the "qualified SUBSAFE activity" representatives, TRFB, access to their work site(s) to verify compliance with SUBSAFE requirements through an initial and follow-on periodic site visits IAW Reference 2.1 Enclosure (1) paragraphs 3 through 3.f and Reference 2.6 Enclosure (1) paragraphs 3 through 3.f.

3.3.4.1. When TRFB has conducted a site visit of the contractor's facility for similar SUBSAFE maintenance within the past twenty-four months from the contract award, follow-on site visits are required prior to the conclusion of the 24 months periodicity requirement IAW Reference 2.1 Enclosure (1) paragraph 3.a and Reference 2.6 Enclosure (1) paragraph 3.a.

3.3.5. Upon award of contract and prior to commencement of any work on SUBSAFE AN/BPS-16 or SUBSAFE AN/BPS-15 radar mast tubes, the contractor shall enter into a SUBSAFE Memorandum of Agreement (MoA) with TRFB to assign responsibilities and actions required by both parties for the conduct of SUBSAFE work under the oversight of TRFB IAW Reference 2.1 Enclosure (1) paragraph 2 and Reference 2.1 Enclosure (1) paragraph 2. The SUBSAFE MoA will not modify or change any requirement imposed herein by the Statement of Work or the awarded contract. The SUBSAFE MoA shall be revised upon award of CLIN(s) added after initial contract award, and prior to commencement of any work on SUBSAFE AN/BPS-16 or SUBSAFE AN/BPS-15 radar mast tube(s) from the additional CLIN(s).

3.3.6. AN/BPS-16 Radar Mast Tube Work Requirements:

3.3.6.1. Reference 2.2 paragraph 3.1.3 requires written records for AN/BPS-16 radar mast tubes in the accomplishment of thermal spray repairs, which are satisfied by Reference 2.2 Enclosure (5); Thermal Spray Job Control Record (TSJCR). The contractor shall provide a completed TSJCR for each radar mast tube; see

Technical Exhibit 1 Deliverables Schedule. If the TSJCR is lost or illegible upon receipt, contact the TRFB COR for a replacement.

- 3.3.6.1.1. Reference 2.2 Enclosure (5) provide a “blank” samples of the TSJCR, and includes instructions on the use of, and completion of, the TSJCR form. Data required to be entered on the TSJCR by the contractor are identified by “(Spray Activity)” and “(Spray Activity) Quality Assurance (QA)”.
- 3.3.6.1.2. The Contractor shall generate and assign a unique Thermal Spray Job Control Record Serial Number (TSJCR S/N) for each AN/BPS-16 radar mast tube IAW Reference 2.2 Enclosure (5). The contractor shall communicate each unique TSJCR S/N to the TRFB COR no later than three workdays following contract award, or upon award of each successive CLIN added after initial contract award. The TSJCR S/N must be expeditiously provided to TRFB to afford time for the application of the TSJCR S/N markings, and list in the SUBSAFE MoA for approvals, by TRFB prior to shipping radar mast tubes to the contractor.
- 3.3.6.2. The contractor shall use and complete the Objective Quality Evidence (OQE) Records, Reference 2.2 applicable Enclosure (6), (7), or (8), during the refurbishment of each AN/BPS-16 radar mast tube IAW Reference 2.2 paragraph 2. The contractor shall provide the completed OQE Record, Reference 2.2 applicable Enclosure (6), (7), or (8), for each mast tube; see Technical Exhibit 1 Deliverables Schedule. If an OQE Record is lost or illegible upon receipt, contact the TRFB COR for a replacement.
- 3.3.6.2.1. The contractor shall use and complete the Outer Diameter Measurement Worksheet, Reference 2.2 Enclosure (3) sheet 5, to document the outer diameter of each radar mast tube during the refurbishment of the AN/BPS-16 radar mast tubes IAW Reference 2.2 Enclosure 1 paragraph 6.a.
- 3.3.6.2.2. The outside diameter of each AN/BPS-16 radar mast tube will be measured and recorded on the Outer Diameter Measurement Worksheet prior to initial machining operations (“PRE-MACHINING (AS RECEIVED) OD”), after machining operations and prior to any application of thermal spray (“POST-MACHINING OD”), and after thermal spray grinding operations and application of sealant (“POST FINISH-GRIND AND SEALANT OD”). Instructions for completing the Outer Diameter Measurement Worksheet are provided by Reference 2.2 Enclosure (3) sheets 6 and 7. If additional guidance for completing the Outer Diameter Measurement Worksheet is required, contact TRFB COR.
- 3.3.6.2.3. The contractor shall provide the three completed Outer Diameter Measurement Worksheets, Reference 2.2 Enclosure (3) sheet 5, for each AN/BPS-16 radar mast tube; see Technical Exhibit 1 Deliverables Schedule. If an Outer Diameter Measurement Worksheet is lost or illegible upon receipt, contact TRFB COR for a replacement.

3.3.6.3. The contractor shall conduct an inspection upon receipt of AN/BPS-16 radar mast tubes prior to commencement of work. The inspection shall confirm receipt of the AN/BPS-16 radar mast tubes with the supplied documentation package. The AN/BPS-16 radar mast tube documentation package is comprised of the following:

3.3.6.3.1. Material Identification (ID) / Control Tag, QA Form 2 – The contractor shall validate the inscribed markings on the AN/BPS-16 radar mast tube correlate to the recorded data in the “MIC NO/SERIAL NO” field of the QA Form 2 attached to the AN/BPS-16 radar mast tube. If a QA Form 2 is lost or illegible upon receipt of a mast tube, notification within twenty-four hours by telephone and email message to the TRFB COR is required.

3.3.6.3.1.1. The QA Form 2 shall remain legible and remain physically attached to the AN/BPS-16 radar mast tube at all times during repair and refurbishment processes, and upon completion of refurbishment the QA Form 2 shall be returned to TRFB physically attached to its respective mast tube.

3.3.6.3.1.2. The contractor shall only remove the QA Form 2 temporarily when it is required to preclude damage during refurbishment processes. When the QA Form 2 requires removal prior to a potentially damaging operation, the QA Form 2 shall be stored in the vicinity of the AN/BPS-16 radar mast tube, such that it cannot be misplaced or damaged, and without delay reattached to the AN/BPS-16 radar mast tube upon the immediate cessation of each potentially damaging process. The contractor shall not make any entry on the QA Form 2; its purpose is for traceability and material control requirements by TRFB instructions.

3.3.6.3.2. Maintenance Certification Record/Re-Entry Control, QA Form 9/QA Form 10 (REC) - The contractor shall validate the “CWP/REC SERIAL NO” recorded in data field “4.” of the REC(s) correlate to the recorded data in the “CWP SER. NO.” field of the QA Form 2 attached to the AN/BPS-16 radar mast tube(s).

3.3.6.3.2.1. The REC(s) shall remain legible, and accompany the associated AN/BPS-16 radar mast tube(s) at all times during repair and refurbishment processes. The REC(s) shall be returned to TRFB with the AN/BPS-16 radar mast tube(s) upon completion of refurbishment. If any REC is lost or illegible upon receipt AN/BPS-16 radar mast tube(s), notification is required within twenty-four hours by telephone and email message to the TRFB COR. The contractor shall not make any entry on any REC; their purpose is for SUBSAFE work authorization and control requirements by TRFB instructions.

3.3.6.3.3. Thermal Spray Job Control Record (TSJCR) – The contractor shall validate the inscribed markings on the AN/BPS-16 radar mast tube

correlate to the recorded data in the "MIC NO." field in Section I of the TSJCR that accompanies each AN/BPS-16 radar mast tube.

- 3.3.6.3.4. Objective Quality Evidence (OQE) Record – The contractor shall verify the OQE Record, Reference 2.2 applicable Enclosures (6), (7), or (8), is enclosed with each AN/BPS-16 radar mast tube.
- 3.3.6.3.5. Outer Diameter Measurement Worksheets – The contractor shall verify receipt of three blank copies of the Outer Diameter Measurement Worksheet, Reference 2.2 Enclosure (3) sheet 5, for documenting "PRE-MACHINING (AS RECEIVED) OD", "POST-MACHINING OD", and "POST FINISH-GRIND AND SEALANT OD", are enclosed with each AN/BPS-16 radar mast tube.
- 3.3.6.3.6. AN/BPS-16 Mast Tube Sketches - The contractor shall verify AN/BPS-16 mast tube sketches, Reference 2.2 Enclosure (3) Sheets 1 through 4, as applicable, are enclosed with each AN/BPS-16 radar mast tube.
- 3.3.6.4. The contractor shall refurbish the outer surfaces of AN/BPS-16 radar mast tubes using thermal spray deposition processes and process controls IAW References 2.2 and Reference 2.3.
 - 3.3.6.4.1. Pre-thermal spray machining of the AN/BPS-16 radar mast tubes shall be performed IAW paragraph 4.4.2 of Reference 2.3.
 - 3.3.6.4.2. The AN/BPS-16 radar mast tube sketches, Enclosure (3) Sheets 1 through 4 of Reference 2.2, provide the process limits and requirements in the "Repair Notes" and "Drawing Notes" blocks and shall be adhered to during radar mast tube pre-thermal spray machining processes.
 - 3.3.6.4.2.1. Each AN/BPS-16 "Rotary Tube" radar mast tube requiring thermal spray restoration shall be performed IAW the supplied radar mast tube sketches, Reference 2.2 Enclosure (3) Sheets 1 through 2. The radar mast tube sketches provide the contractor with the TRFB Engineering authorization of the specified repair areas of the "Rotary Tube" radar mast tube and shall be adhered to during pre-thermal spray machining processes.
 - 3.3.6.4.2.2. The base material and yield strength properties of AN/BPS-16 radar mast tubes are provided in paragraph 5 of Enclosure (2) of Reference 2.2.
 - 3.3.6.4.3. NDT PT meeting the requirements of Reference 2.5 shall be conducted upon completion of pre-thermal spray machining operations. Ensure NDT PT notification(s) comply with SOW paragraph 3.3.3.2.1.
 - 3.3.6.4.4. Reference 2.2 Enclosure (2) paragraph 6.a requires abrasive blasting be performed IAW paragraph 4.4.2.5 of Reference 2.3.
 - 3.3.6.4.5. Reference 2.2 Enclosure (2) paragraph 6.a requires solvent cleaning be performed IAW paragraph 4.4.1 of Reference 2.3.

3.3.6.4.6. Reference 2.2 Enclosure (2) paragraph 3 requires the type of thermal spray as plasma spray using a Monel (NiCu) bond coat with an Alumina-Titania finish coat, or a Nickel-Aluminum-Molybdenum (NiAlMo) bond coat with Alumina-Titania finish coat.

3.3.6.4.6.1. Reference 2.2 Enclosure (2) paragraph 4 specifies the maximum thickness for the thermal spray Monel (NiCu) or Nickel-Aluminum-Molybdenum (NiAlMo) bond coat and Alumina-Titania finish coat.

3.3.6.4.7. Reference 2.2 Enclosure (2) paragraph 6.b.(1) through 6.b.(3)(i) specifies the tolerance for outside diameter and surface finish of the thermal spray restoration of AN/BPS-16 radar mast tubes.

3.3.6.4.8. Reference 2.2 Enclosure (2) paragraph 7 requires inspection of thermal sprayed coatings IAW paragraph 3.5.2.1 of Reference 2.3.

3.3.6.5. The contractor shall ensure cleanliness of the AN/BPS-16 radar mast tubes prior to their return to TRFB. Radar mast tubes shall be in a similar state of cleanliness as when received from TRFB and be free of industrial residue and debris from the thermal spray processes.

3.3.6.6. During the TRFB review of each AN/BPS-16 radar mast tube's submitted TSJCR, OQE Record, and Outer Diameter Measurement Worksheet documentation, if TRFB determines corrections/changes are required to meet the requirements of Reference 2.2 for records retention, the contractor shall make corrections/changes, or provide justification to not correct/change, within three workdays of notification by email message.

3.3.6.7. When SUBSAFE AN/BPS-16 radar mast tubes are at the contractor's facility for more than three workdays, the contractor shall provide weekly progress reports no later than close of business Wednesday to the TRFB COR concerning work in process. The content of the weekly progress report shall express completion percentages for each radar mast tube, completion projections for the following week, and the proposed dates for performance of NDT PT. Ensure NDT PT notifications comply with SOW paragraph 3.3.3.2.1.

3.3.6.8. The contractor shall identify any discrepancies discovered during receipt of, or work in process of, AN/BPS-16 radar mast tubes. Communication of the discrepancies shall be made to the TRFB COR not later than twenty-four hours of discrepancy discovery.

3.3.7. AN/BPS-15 Radar Mast Tube Work Requirements:

3.3.7.1. Reference 2.3 paragraph 3.1.3 requires written records for AN/BPS-15 radar mast tubes in the accomplishment of thermal spray repairs, which are satisfied by Reference 2.1 Enclosure (5); Thermal Spray Job Control Record (TSJCR). The contractor shall provide a completed TSJCR for each radar mast tube; see Technical Exhibit 1 Deliverables Schedule. If the TSJCR is lost or illegible upon receipt, contact TRFB COR for a replacement.

- 3.3.7.1.1. Reference 2.1 Enclosure (5) provides a “blank” sample of the TSJCR, and includes instructions on the use of, and completion of, the TSJCR. Data required to be entered on the TSJCR by the contractor are identified by “(Spray Activity)” and “(Spray Activity) Quality Assurance (QA)”.
- 3.3.7.1.2. The Contractor shall generate and assign a unique thermal spray serial number for each AN/BPS-15 radar mast tube IAW Reference 2.1 Enclosure (5). The contractor shall communicate the unique thermal spray serial numbers to the TRFB COR no later than three workdays following contract award, or upon award of each successive CLIN added after initial contract award. The TSJCR S/N must be expeditiously provided to TRFB to afford time for the application of the TSJCR S/N markings, and list in the SUBSAFE MoA for approvals, by TRFB prior to shipping radar mast tubes to the contractor.
- 3.3.7.2. The contractor shall use and complete the Objective Quality Evidence (OQE) Records, Reference 2.1 applicable Enclosure (6), (7), or (8), during the refurbishment of the AN/BPS-15 radar mast tubes IAW Reference 2.1 paragraph 2. The contractor shall provide the completed OQE Record, Reference 2.1 applicable Enclosures (6), (7), or (8), for each radar mast tube; see Technical Exhibit 1 Deliverables Schedule. If an OQE Record is lost or illegible upon receipt, contact TRFB COR for a replacement.
- 3.3.7.2.1. The contractor shall use and complete the Outer Diameter Measurement Worksheet, Reference 2.1 Enclosure (3) sheet 5, to document the outer diameter of each radar mast tube during the refurbishment of the AN/BPS-15 radar mast tubes IAW Reference 2.1 Enclosure 1 paragraph 7.a.
- 3.3.7.2.2. The outside diameter of each AN/BPS-15 radar mast tube will be measured and recorded on the Outer Diameter Measurement Worksheet prior to initial machining operations (“PRE-MACHINING (AS RECEIVED) OD”), after machining operations and prior to any application of thermal spray (“POST-MACHINING OD”), and after thermal spray grinding operations and application of sealant (“POST FINISH-GRIND AND SEALANT OD”). Instructions for completing the Outer Diameter Measurement Worksheet are provided by Reference 2.1 Enclosure (3) sheets 6 and 7. If additional guidance for completing the Outer Diameter Measurement Worksheet is required, contact TRFB COR.
- 3.3.7.2.3. The contractor shall provide the three completed Outer Diameter Measurement Worksheets, Reference 2.1 Enclosure (3) sheet 7, for each AN/BPS-15 radar mast tube; see Technical Exhibit 1 Deliverables Schedule. If an Outer Diameter Measurement Worksheet is lost or illegible upon receipt, contact the TRFB COR for a replacement.
- 3.3.7.3. The contractor shall conduct an inspection upon receipt of an AN/BPS-15 radar mast tube prior to commencement of work. The inspection shall confirm receipt of the AN/BPS-15 radar mast tube with the supplied documentation

package. The AN/BPS-15 radar mast tube documentation package is comprised of the following:

- 3.3.7.3.1. Material Identification (ID) / Control Tag, QA Form 2 – The contractor shall validate the inscribed markings on the AN/BPS-15 radar mast tube correlate to the recorded data in the “MIC NO/SERIAL NO” field of the QA Form 2 attached to the AN/BPS-15 radar mast tube. If a QA Form 2 is lost or illegible upon receipt of an AN/BPS-15 radar mast tube, notification, within twenty-four hours, by telephone and email message to the TRFB COR is required.
 - 3.3.7.3.1.1. The QA Form 2 shall remain legible and remain physically attached to the AN/BPS-15 radar mast tube at all times during repair and refurbishment processes and upon completion of refurbishment the QA Form 2 shall be returned to TRFB physically attached to the AN/BPS-15 radar mast tube.
 - 3.3.7.3.1.2. The contractor shall only remove the QA Form 2 temporarily when it is required to preclude damage during refurbishment processes. When the QA Form 2 requires removal prior to a potentially damaging operation, the QA Form 2 shall be stored in the vicinity of the AN/BPS-15 radar mast tube such that it cannot be misplaced or damaged, and without delay reattached to the AN/BPS-15 radar mast tube upon the immediate cessation of each potentially damaging process. The contractor shall not make any entry on the QA Form 2; its purpose is for traceability and material control requirements by TRFB instructions.
- 3.3.7.3.2. Maintenance Certification Record/Re-Entry Control, QA Form 9/QA Form 10 (REC) - The contractor shall validate the “CWP/REC SERIAL NO” recorded in data field “4.” of the REC correlates to the recorded data in the “CWP SER. NO.” field of the QA Form 2 attached to the AN/BPS-15 radar mast tube.
 - 3.3.7.3.2.1. The REC shall remain legible, and accompany the AN/BPS-15 radar mast tube at all times during repair and refurbishment processes. The REC shall be returned to TRFB with the AN/BPS-15 radar mast tube upon completion of refurbishment. If a REC is lost or illegible upon receipt of AN/BPS-15 radar mast tube, notification, within twenty-four hours, by telephone and email message to the TRFB COR is required. The contractor shall not make any entry on the REC; its purpose is for SUBSAFE work authorization and control requirements by TRFB instructions.
- 3.3.7.3.3. Thermal Spray Job Control Record (TSJCR) – The contractor shall validate the inscribed markings on the AN/BPS-15 radar mast tube correlate to the recorded data in the “MIC NO.” field in Section I of the TSJCR that accompanies the AN/BPS-15 radar mast tube.

- 3.3.7.3.4. Objective Quality Evidence (OQE) Record – The contractor shall verify the OQE Record, Reference 2.1 applicable Enclosures (9) or (10), is enclosed with the AN/BPS-15 radar mast tube.
- 3.3.7.3.5. Outer Diameter Measurement Worksheets – The contractor shall verify receipt of three blank copies of the Outer Diameter Measurement Worksheet, Reference 2.1 Enclosure (3) sheet 5, for documenting “PRE-MACHINING (AS RECEIVED) OD”, “POST-MACHINING OD”, and “POST FINISH-GRIND AND SEALANT OD”, are enclosed with each AN/BPS-16 radar mast tube.
- 3.3.7.3.6. AN/BPS-15 Mast Tube Sketches - The contractor shall verify AN/BPS-15 radar mast tube sketches, Reference 2.1 Enclosure (3) applicable sheets 1 through 4, are enclosed with each AN/BPS-15 radar mast tube.
- 3.3.7.4. The contractor shall refurbish the outer surfaces of both “wear and non-wear areas” of the AN/BPS-15 Radar mast tube using thermal spray deposition processes and process controls IAW References 2.1 and Reference 2.3.
- 3.3.7.4.1. Pre-thermal spray machining of the AN/BPS-15 Radar mast tube “wear and non-wear area” shall be performed IAW paragraph 4.4.2 of Reference 2.3.
- 3.3.7.4.1.1. The AN/BPS-15 Radar mast tube “wear and non-wear area” sketch, Enclosure (3) of Reference 2.1, provide additional process limits and requirements in the “Repair Notes” and “Drawing Notes” blocks that shall be adhered to during pre-thermal spray machining processes.
- 3.3.7.4.1.2. The base material and yield strength properties of the AN/BPS-15 Radar mast tube are provided in Reference 2.1 Enclosure (2) paragraph 7.
- 3.3.7.4.2. Nondestructive testing PT meeting the requirements of Reference 2.5 shall be conducted upon completion of pre-thermal spray machining operations. Ensure NDT PT notifications(s) comply with SOW paragraph 3.3.3.2.1.
- 3.3.7.4.3. Reference 2.1 Enclosure (2) paragraph 8.a requires solvent cleaning be performed IAW paragraph 4.4.1. of Reference 2.3.
- 3.3.7.4.4. Reference 2.1 Enclosure (2) paragraph 8.a requires abrasive blasting be performed IAW paragraph 4.4.2.5 of Reference 2.3.
- 3.3.7.4.5. Reference 2.1 Enclosure (2) paragraph 5 requires the type of thermal spray as plasma spray using a Monel (NiCu) bond coat and Alumina-Titania finish coat, or a Nickel-Aluminum-Molybdenum (NiAlMo) bond coat with Alumina-Titania finish coat.
- 3.3.7.4.5.1. Reference 2.1 Enclosure (2) paragraph 6 specifies the maximum thickness for the thermal spray Monel (NiCu) or Nickel-

Aluminum-Molybdenum (NiAlMo) bond coat and Alumina-Titania finish coat.

- 3.3.7.4.6. Reference 2.1 Enclosure (2) paragraphs 8.b.(1)(i) and 8.b.(1)(ii) specifies the tolerance for outside diameter and surface finish of the thermal spray restoration of the AN/BPS-15 Radar mast tubes.
- 3.3.7.4.7. Reference 2.1 Enclosure (2) paragraph 9 requires inspection of thermal sprayed coatings IAW paragraph 3.5.2.1 of Reference 2.3.
- 3.3.7.5. The contractor shall ensure cleanliness of the AN/BPS-15 radar mast tube prior to its return to TRFB. Mast tubes shall be in a similar state of cleanliness as when received from TRFB and shall be free of industrial residue and debris from the thermal spray processes.
- 3.3.7.6. During the TRFB review of each AN/BPS-15 radar mast tube's submitted TSJCR and/or OQE Record, if TRFB determines correction/changes are required to meet the requirements of Reference 2.1 for records retention, the contractor shall make corrections/changes within three workdays of notification by email message.
- 3.3.7.7. When SUBSAFE AN/BPS-15 radar mast tubes are at the contractor's facility for more than three workdays, the contractor shall provide weekly progress reports no later than close of business on every Wednesday to the TRFB COR concerning work in process. The content of the weekly progress report shall express completion percentages for each radar mast tube, completion projections for the following week, and the proposed dates for performance of NDT PT. Ensure NDT PT notifications(s) comply with SOW paragraph 3.3.3.2.1.
- 3.3.7.8. The contractor shall identify any discrepancies discovered during receipt of, or work in process of, the AN/BPS-15 radar mast tube. Communication of the discrepancies shall be made to the TRFB COR not later than twenty-four hours of discrepancy discovery.

4. GOVERNMENT REQUIREMENTS:

- 4.1. TRFB nondestructive testing division, code 630, NDT Level III Examiner will evaluate the contractor's, or sub-contractor's if used, submitted records in accordance with SOW paragraph 3.3.3 to ensure compliance with all NAVSEA requirements for NDT PT, to include Reference 2.5 and Reference 2.6.
 - 4.1.1. TRFB nondestructive testing division, code 630, NDT Level III Examiner may determine a need to assess the performance of NDT PT by the contractor, or subcontractor, to verify their compliance with the requirements of Reference 2.5 and Reference 2.6. This assessment, NDT PT performance review, when necessary, will be performed by initial, and potential follow-on, site visit(s) by TRFB nondestructive testing division, code 630, personnel.
 - 4.1.1.1. The TRFB nondestructive testing division, code 630, personnel performing the NDT PT performance review are permitted to issue a stoppage of NDT PT

work, for not more than forty-eight hours, when NDT PT certification discrepancies are observed. Communication of work stoppage and the NDT PT certification discrepancies will be made to the TRFB COR within four hours of work stoppage.

- 4.1.2. Site visits of the contractor's facility for the initial, and potential follow-on, NDT PT performance review will be coordinated by the TRFB COR with input from the contractor concerning the period of performance (dates and schedule) of site visits.

- 4.1.2.1. The funding and the arrangement of travel of the TRFB employees for the purpose of initial and follow-on site visit(s) is the responsibility of TRFB.

- 4.1.2.2. TRFB engineering division, code 410, and SUBSAFE Program division, code 06, personnel will conduct the initial assessment, and any assessment(s) required forthwith every twenty-four months, to verify the contractor's compliance with SUBSAFE requirements IAW Reference 2.2 Enclosure (1) paragraphs 3 through 3.f and Reference 2.1 Enclosure (1) paragraphs 3 through 3.f.

- 4.1.3. Site visits of the contractor's facility for the initial assessment, or follow-on review, will be coordinated by the TRFB COR with input from the contractor concerning the period of performance (dates and schedule) of site visits.

- 4.1.3.1. The funding and the arrangement of travel of the TRFB employees for the purpose of initial and follow-on site visit(s) is the responsibility of TRFB.

- 4.2. Upon award of contract TRFB will enter into a SUBSAFE Memorandum of Agreement (MOA) with the contractor to assign responsibilities and actions required by both parties for the conduct of SUBSAFE work under the oversight of TRFB IAW Reference 2.2 Enclosure (1) paragraph 2 and Reference 2.1 Enclosure (1) paragraph 2. The SUBSAFE MoA will be revised upon award of CLIN(s) added after initial contract award, and prior to shipping any radar mast tube(s) from the additional CLIN(s).

4.3. AN/BPS-16 Radar Mast Tube Government (TRFB) Requirements:

- 4.3.1. For each AN/BPS-16 radar mast tube, TRFB will provide the TSJCR form, Reference 2.2 Enclosure (5), within the shipping crate, which is pre-filled by TRFB IAW Reference 2.2 Enclosure (5).

- 4.3.1.1. TRFB will apply the unique thermal spray serial number to each AN/BPS-16 radar mast tube IAW Reference 2.2 Enclosure (1) paragraph 4.d. in the specified area IAW Reference 2.2 Enclosure (3).

- 4.3.2. TRFB will provide the OQE Record, Reference 2.2 applicable Enclosures (6), (7), or (8), for each AN/BPS-16 radar mast tube within the shipping crate, for use during refurbishment.

- 4.3.3. TRFB will provide three blank Outer Diameter Measurement Worksheets, Reference 2.2 Enclosure (3) sheet 4, for each AN/BPS-16 radar mast tube within the shipping crate, for use during refurbishment.

4.3.4. TRFB will provide the REC(s), within the shipping crate, prior to shipping AN/BPS-16 radar mast tubes to the contractor.

4.3.5. TRFB will physically attach the Material Identification (ID) / Control Tag, QA Form 2, to each AN/BPS-16 radar mast tube prior to shipping mast tubes to the contractor.

4.3.6. A review of each AN/BPS-16 radar mast tube's submitted TSJCR, OQE Record and Outer Diameter Measurement Worksheet documentation will be performed by TRFB engineering division code 413, and TRFB material certification branch code 612, prior to records acceptance and retention.

4.3.7. During review, if TRFB determines correction/changes are required to meet the record requirements of Reference 2.2, notification to the contractor will be made via the COR within seven workdays of receipt of the AN/BPS-16 radar mast tubes' "electronically" submitted TSJCR, OQE Record and Outer Diameter Measurement Worksheet documentation.

4.3.8. TRFB shall perform a technical receipt inspection of each restored AN/BPS-16 radar mast tube upon receipt from the contractor IAW Reference 2.2 Enclosure (1) paragraph 8 through 8.d.

4.3.9. The completion of inspections, review of documentation, and acceptability of the contractor's services will be completed within ten workdays from the date of receipt of the restored AN/BPS-16 radar mast tubes.

4.4. AN/BPS-15 Radar Mast Tube Government (TRFB) Requirements:

4.4.1. For each AN/BPS-15 radar mast tube, TRFB will provide the TSJCR, Reference 2.1 Enclosure (5), within the shipping crate, which is pre-filled by TRFB IAW Reference 2.1 Enclosure (5).

4.4.1.1. TRFB will apply the unique thermal spray serial number to each AN/BPS-15 radar mast tube IAW Reference 2.1 Enclosure (1) paragraph 5.c. in the specified area IAW Reference 2.1 Enclosure (3).

4.4.1.2. TRFB will provide the OQE Record, Reference 2.1 applicable Enclosures (9) or (10), with each AN/BPS-15 radar mast tube, within the shipping crate, for use during refurbishment.

4.4.1.3. TRFB will provide the REC, within the shipping crate, prior to shipping an AN/BPS-15 radar mast tube to the contractor.

4.4.1.4. TRFB will physically attach the Material Identification (ID) / Control Tag, QA Form 2, to each AN/BPS-15 radar mast tube prior to shipping a radar mast tube to the contractor.

4.4.1.5. A review of each AN/BPS-15 radar mast tube's submitted TSJCR, OQE Record and Outer Diameter Measurement Worksheet documentation will be performed by TRFB engineering division, code 413, and TRFB material certification branch, code 612, prior to acceptance and retention.

4.4.1.5.1. During review, if TRFB determines changes are required to meet the record requirements of Reference 2.1 notification to the contractor will be made via the COR within seven (7) workdays of receipt of the AN/BPS-15 radar mast tube's "electronically" submitted TSJCR and OQE Record.

4.4.2. TRFB shall perform a technical receipt inspection of each restored AN/BPS-15 radar mast tube upon receipt from the contractor IAW Reference 2.1 Enclosure (1) paragraph 9 through 9.d.

4.4.3. The completion of inspections, review of documentation, and acceptability of the contractor's services will be completed within ten workdays from the date of receipt of each restored AN/BPS-15 radar mast tube.

5. **CONTRACTOR FURNISHED MATERIALS:** The Contractor shall provide all materials (consumable or otherwise) required in the performance of contracted work at the contractor's facility.
6. **QUALITY:** The Quality Assurance Surveillance Plan (QASP) will be used as a tool to verify the contractor is performing all services and delivery/installation of replacement parts required by the above requirements in a timely, accurate and complete fashion.
7. **DELIVERABLE ITEMS:** Furnish deliverables per the following Technical Exhibit 1 Deliverables Schedule:

Technical Exhibit 1 Deliverables Schedule

<u>Deliverable</u>	<u>Frequency</u>	<u>Quantity</u>	<u>Medium</u>	<u>Recipient</u>
NAVSEA Approval of QA Program per paragraph 3.3.1.	Once, provide with contract solicitation proposal	1 copy	Paper/PDF	Contracting Officer
NAVSEA Approval of Thermal Spray Facility; per paragraph 3.3.1.	Once, provide with contract solicitation proposal	1 copy	Paper/PDF	Contracting Officer
NAVSEA Approval of Thermal Spray Procedures per paragraphs 3.3.1.	Once, provide with contract solicitation proposal	1 copy	Paper/PDF	Contracting Officer
NDT Vendor Qualification Documentation per paragraphs 3.3.3 to 3.3.3.1.6.	Once, provide with contract solicitation proposal	1 copy	Paper/PDF	Contracting Officer

Thermal Spray Job Control Record (TSJCR) per paragraphs 3.3.6.1. and 3.3.7.1.	Multiple, at work completion for each AN/BPS-15 or AN/BPS-16 Mast Tube.	Original (enclosed with GFE item); 1 copy (emailed prior to crating GFE for return)	Paper (original) and PDF (copy)	Enclosed with GFE (original) and COR (copy)
Objective Quality Evidence (OQE) Record per paragraphs 3.3.6.2. and 3.3.7.2.	Multiple, at work completion for each AN/BPS-15 or AN/BPS-16 Mast Tube.	Original (enclosed with GFE item); 1 copy (emailed prior to crating GFE for return)	Paper (original) and PDF (copy)	Enclosed with GFE (original) and COR (copy)
Outer Diameter Measurement Worksheet: PRE-MACHINE (AS RECEIVED) OD per paragraphs 3.3.6.2.1 to 3.3.6.2.3 and 3.3.7.2.1 to 3.3.7.2.3.	Multiple, at work completion for each AN/BPS-15 or AN/BPS-16 Mast Tube.	Original (enclosed with GFE item); 1 copy (emailed prior to crating GFE for return)	Paper (original) and PDF (copy)	Enclosed with GFE (original) and COR (copy)
Outer Diameter Measurement Worksheet; POST-MACHINING OD per paragraphs 3.3.6.2.1 to 3.3.6.2.3 and 3.3.7.2.1 to 3.3.7.2.3.	Multiple, at work completion for each AN/BPS-15 or AN/BPS-16 Mast Tube.	Original (enclosed with GFE item); 1 copy (emailed prior to crating GFE for return)	Paper (original) and PDF (copy)	Enclosed with GFE (original) and COR (copy)
Outer Diameter Measurement Worksheet; POST FINISH-GRIND AND SEALANT OD per paragraphs 3.3.6.2.1 to 3.3.6.2.3 and 3.3.7.2.1 to 3.3.7.2.3.	Multiple, at work completion for each AN/BPS-15 or AN/BPS-16 Mast Tube.	Original (enclosed with GFE item); 1 copy (emailed prior to crating GFE for return)	Paper (original) and PDF (copy)	Enclosed with GFE (original) and COR (copy)